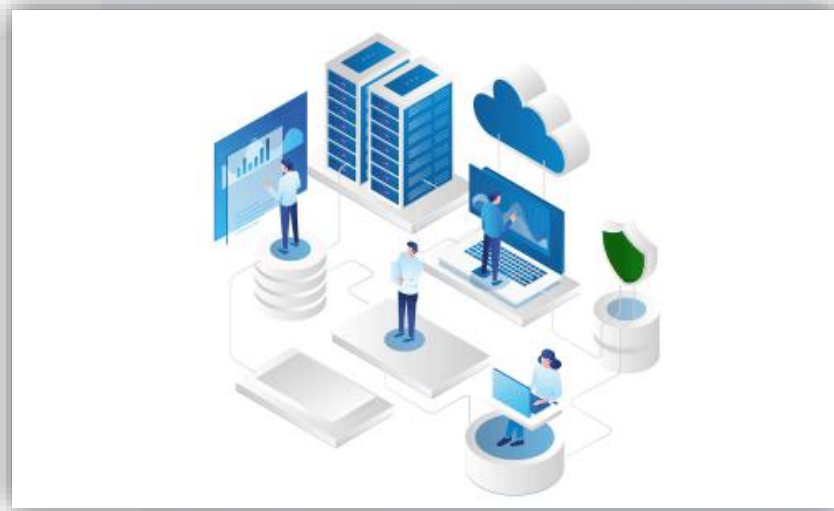


Introduction to Data Engineering

The Role, Key Concepts, and Tools of Data Engineers



Lecture by: MR. DINA CHANTHAN

Contents

1. Learning Objectives
2. What is Data Engineering?
3. Role of Data Engineer
4. Key Concepts of Data Engineering
5. Data Pipelines
6. Batch vs Real-Time Processing

Contents (Cont.,)

7. **Big Data**
8. **Popular Data Engineering Tools**
9. **The Data Engineering Lifecycle**
10. **Collaboration in Data Engineering**
11. **Summary**
12. **Hand-On Labs**
13. **Homework Assignment**

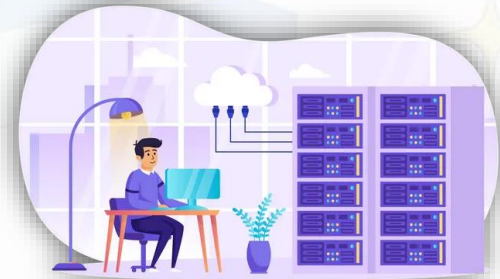
1. Learning Objectives

➤ **By the end of this session, you will**

- Understand the role and responsibilities of data engineer.
- Grasp the basic concepts of data engineering, include ETL, data pipelines, and big data.
- Learn about key tools and technologies used in data engineering.

2. What is Data Engineering?

- **Definition:** Data engineering is the process of involves designing, building, and maintaining the infrastructure and tools to collect, store, and process data at scale.
- **Importance:** Enables data-driven decision-making for businesses.
- **Example:** Thinking about how Netflix recommends movies based on your preferences. Data engineers ensure the data pipeline behind the scenes runs smoothly to deliver accurate recommendations.



3. Role of Data Engineer

➤ Core Responsibilities:

- Build and maintain data pipelines.
- Ensure the data infrastructure is scalable and efficient.
- Work with data scientists and analysts.

➤ Analogy:

- Data engineers are like architects for data infrastructure.

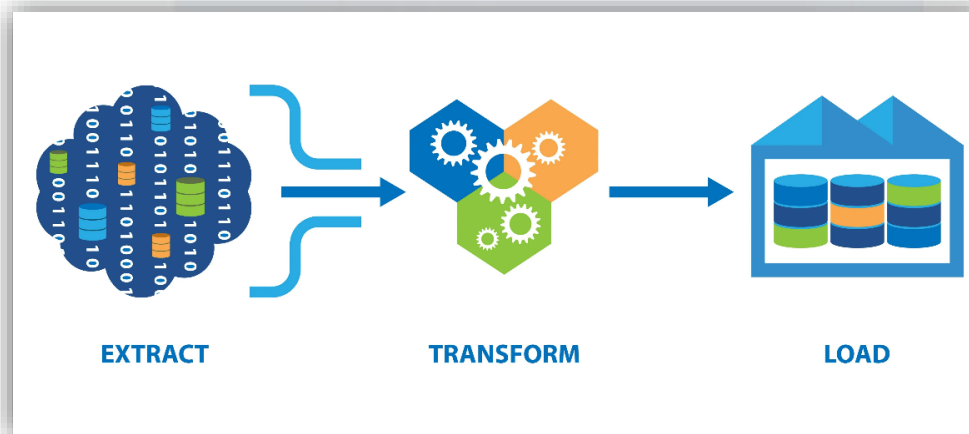
4. Key Concepts in Data Engineering

➤ ETL (Extract, Transform, Load):

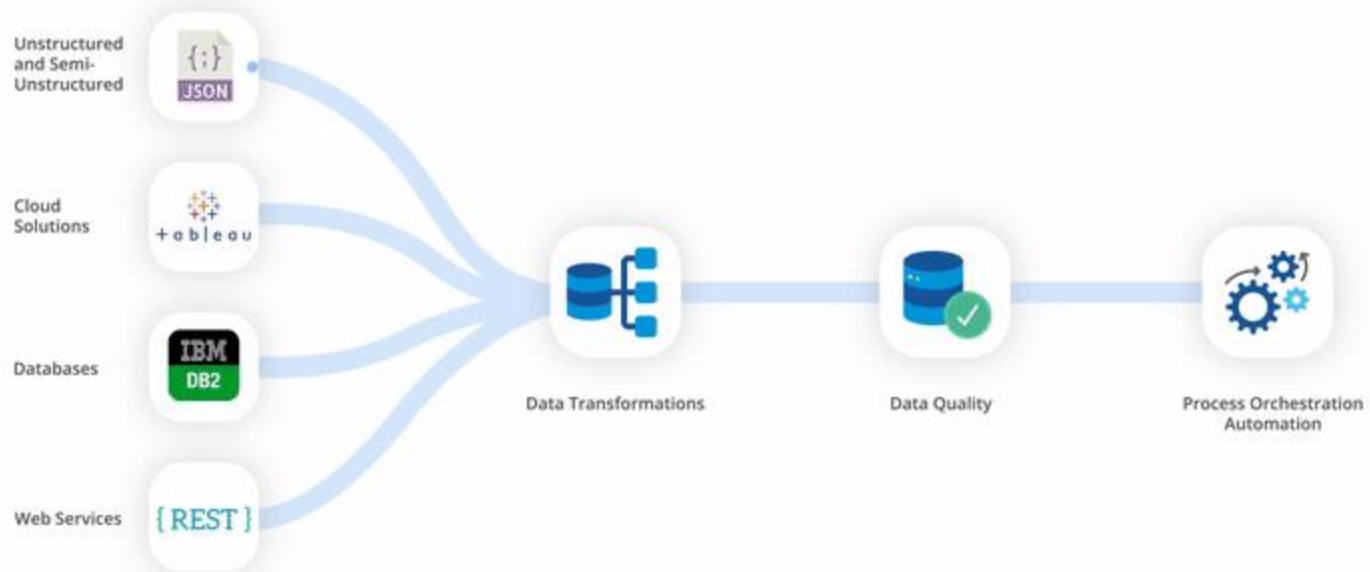
- Extract: Gather data from various sources.
- Transform: Clean and process data.
- Load: Store data in a data warehouse.

➤ Example:

- How Amazon process millions of orders.



ETL- Extract, Transform, and Load

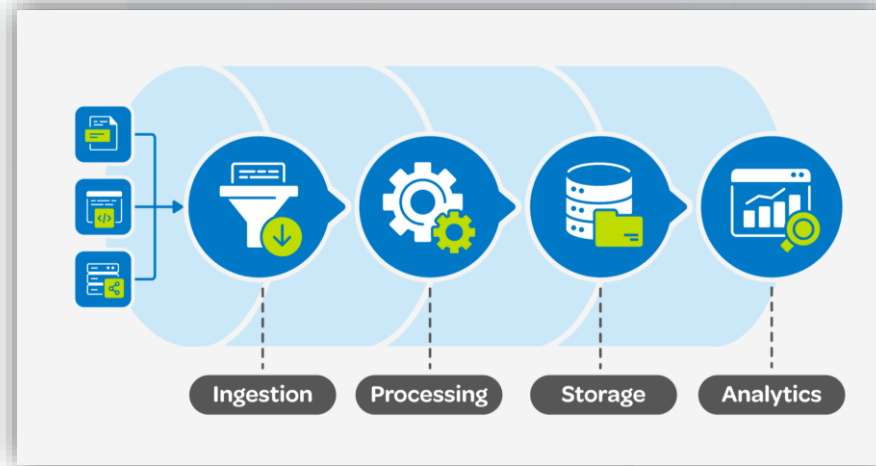


5. Data Pipelines

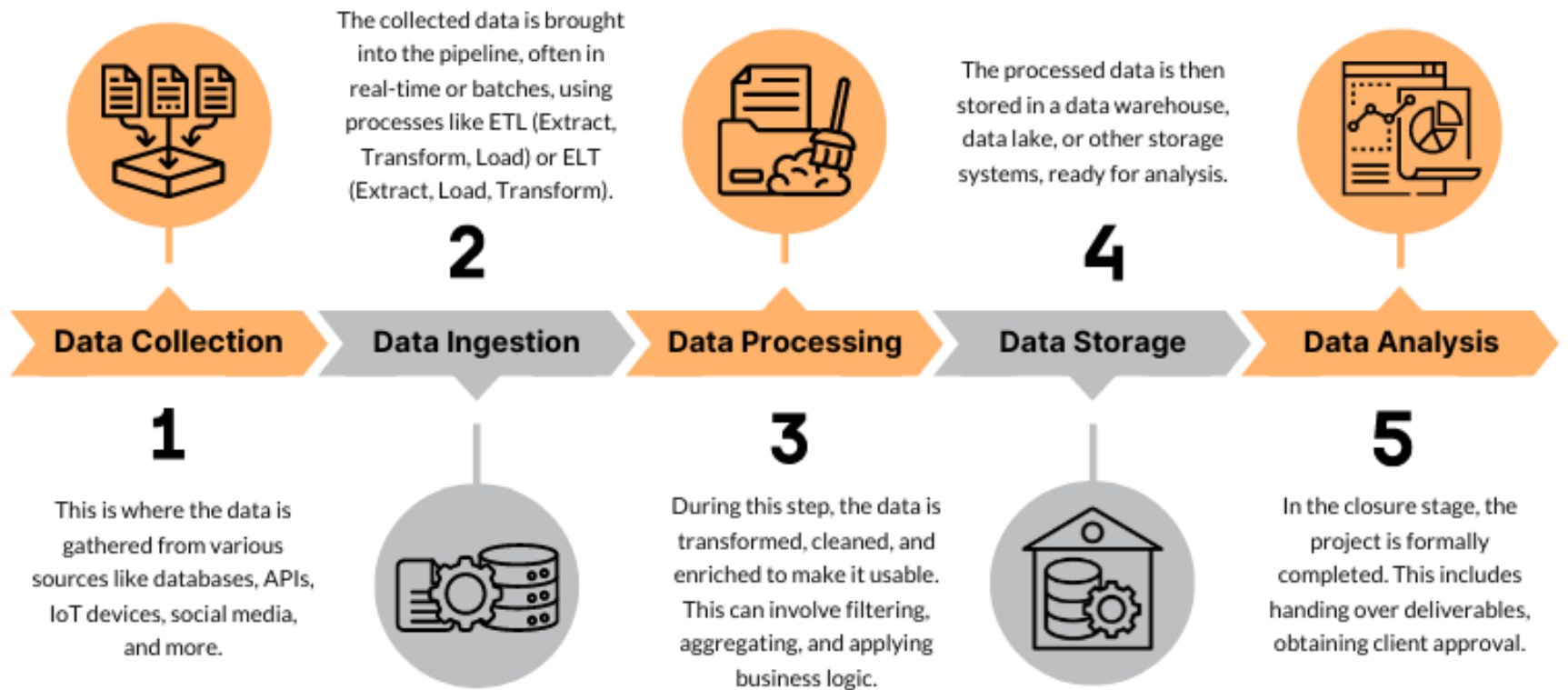
➤ **Definition:** A set of automated workflows that move data from one place to another.

➤ **Example:**

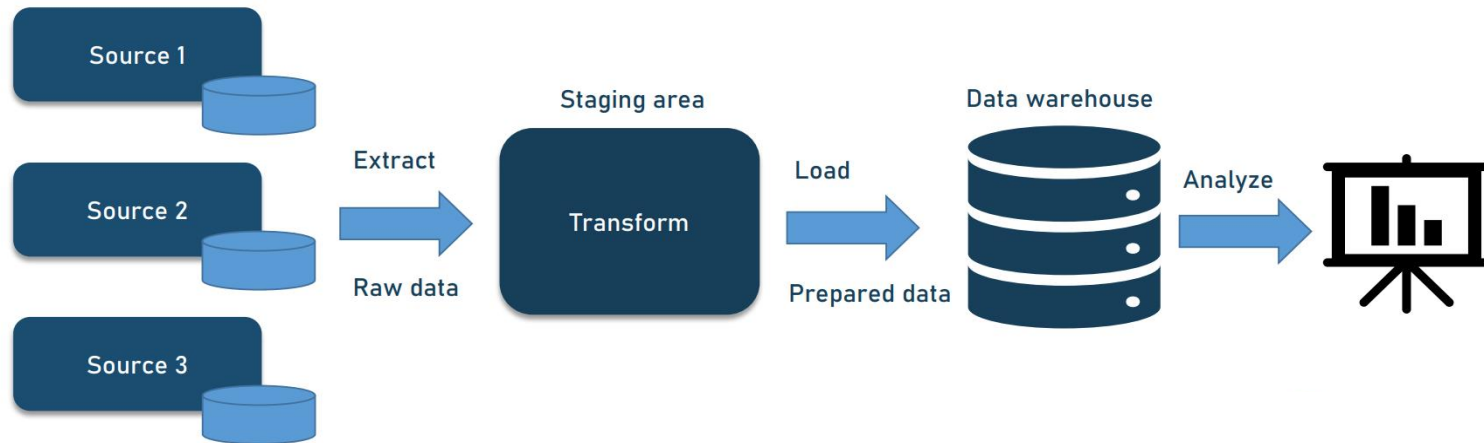
- Think of pipelines as the plumbing of data – ensuring the right data flows to the right place at the right time.



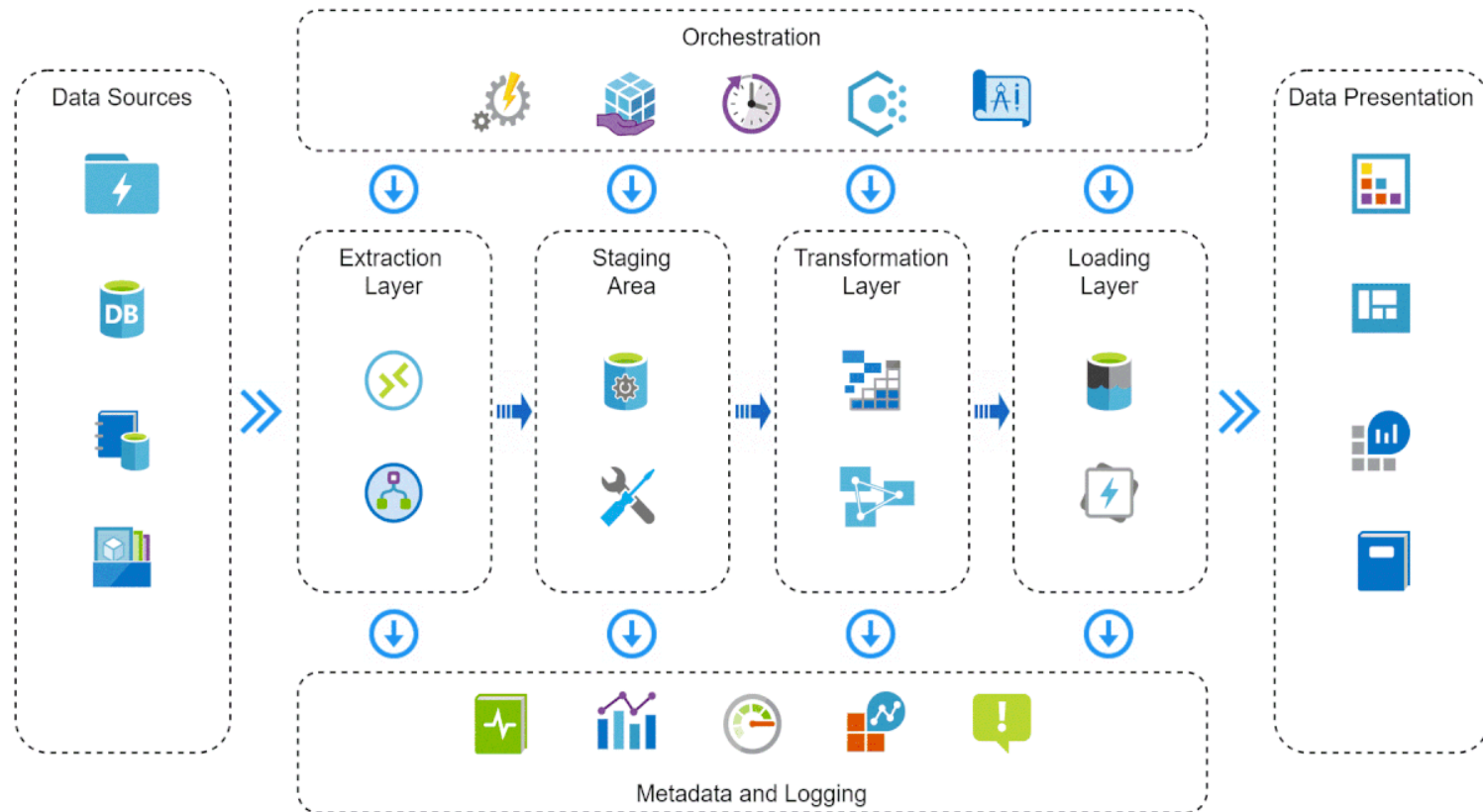
Data Pipelines



ETL Pipelines



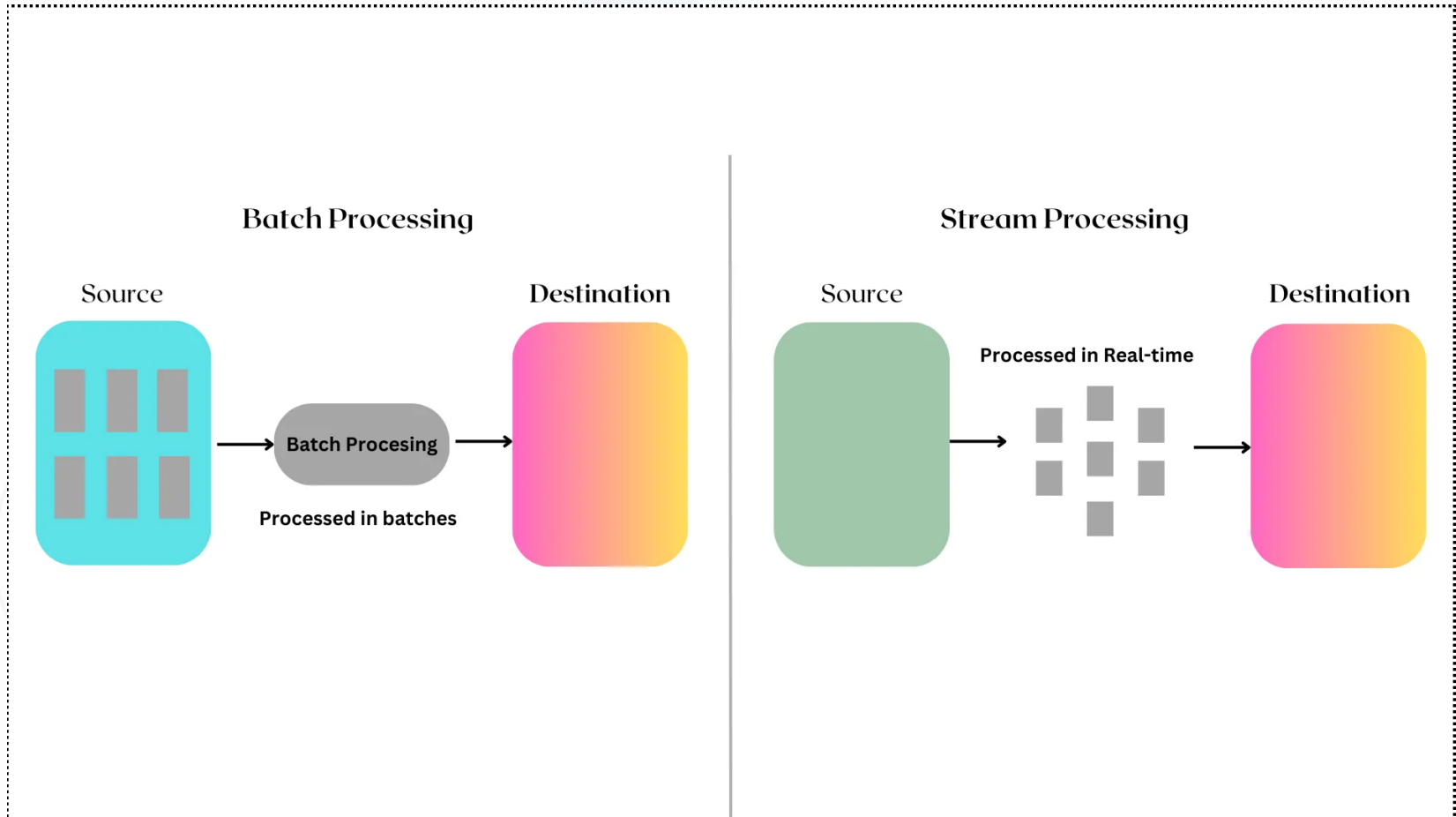
Pipelines Architectures



6. Batch vs Real-Time Processing

- **Batch Processing:** Processes large volumes of data in chunks (e.g., daily reports).
- **Real-Time Processing:** Processes data continuously as it arrives (e.g., livestock prices)
- **Example:** Comparing how banks process transactions overnight (batch) vs how twitter show live tweets (real-time).

Batch vs Real-Time Processing



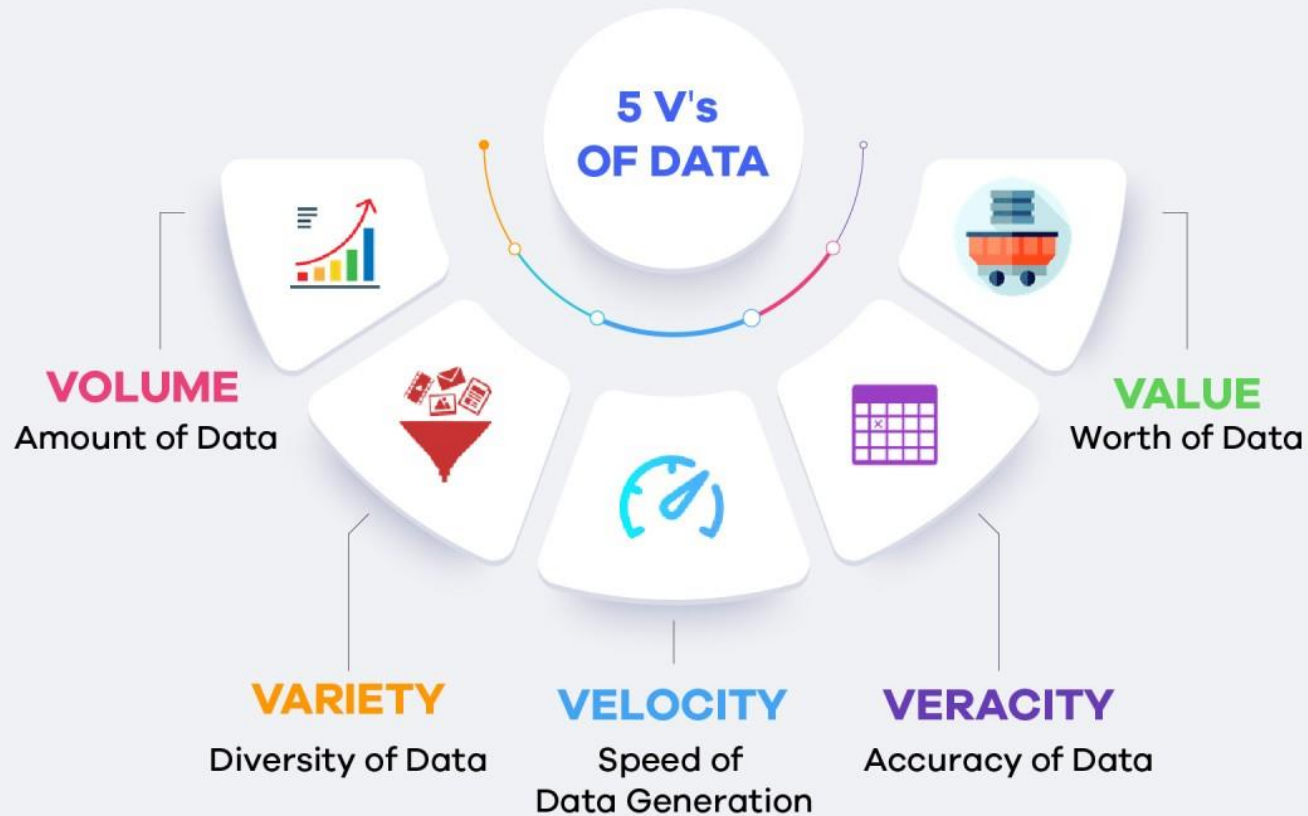
Batch vs Real-Time Processing

	Batch processing	Stream processing
Frequency	Infrequent jobs that produce results once the job has finished running	Continuously running jobs that produce constant results
Performance	High latency (minutes to hours)	Low latency (milliseconds to seconds)
Data sources	Databases, APIs, statics files	Message queues, event streams, clickstreams
Analysis type	Complex analysis, ideal for very large data sets and projects that involve deeper data analysis	Relatively simpler analysis, including aggregation, filtering, enrichment, event detection, etc.
Processing	Collects data over time and sends it for processing once collected	Continuously collects data and processes it fast, piece by piece
Use case	Generally meant for large data quantities that are not time sensitive	Typically meant for data needed immediately

- **Definition:** Handling and processing data sets too large for traditional systems.
- **Example:** Google processing billions of search queries daily.
- **Tools:** Introduce Hadoop for managing big data.

- [illegible]

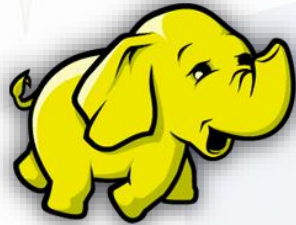
5V's of Big Data



8. Popular Data Engineering Tools

- **Hadoop:** Distributed storage and processing of big data.
- **Apache Spark:** Fast, distributed processing of large datasets.
- **Apache Kafka:** Real-time data streaming.
- **Apache Airflow:** Orchestration of workflows.

Popular Data Engineering Tools



Apache
Airflow



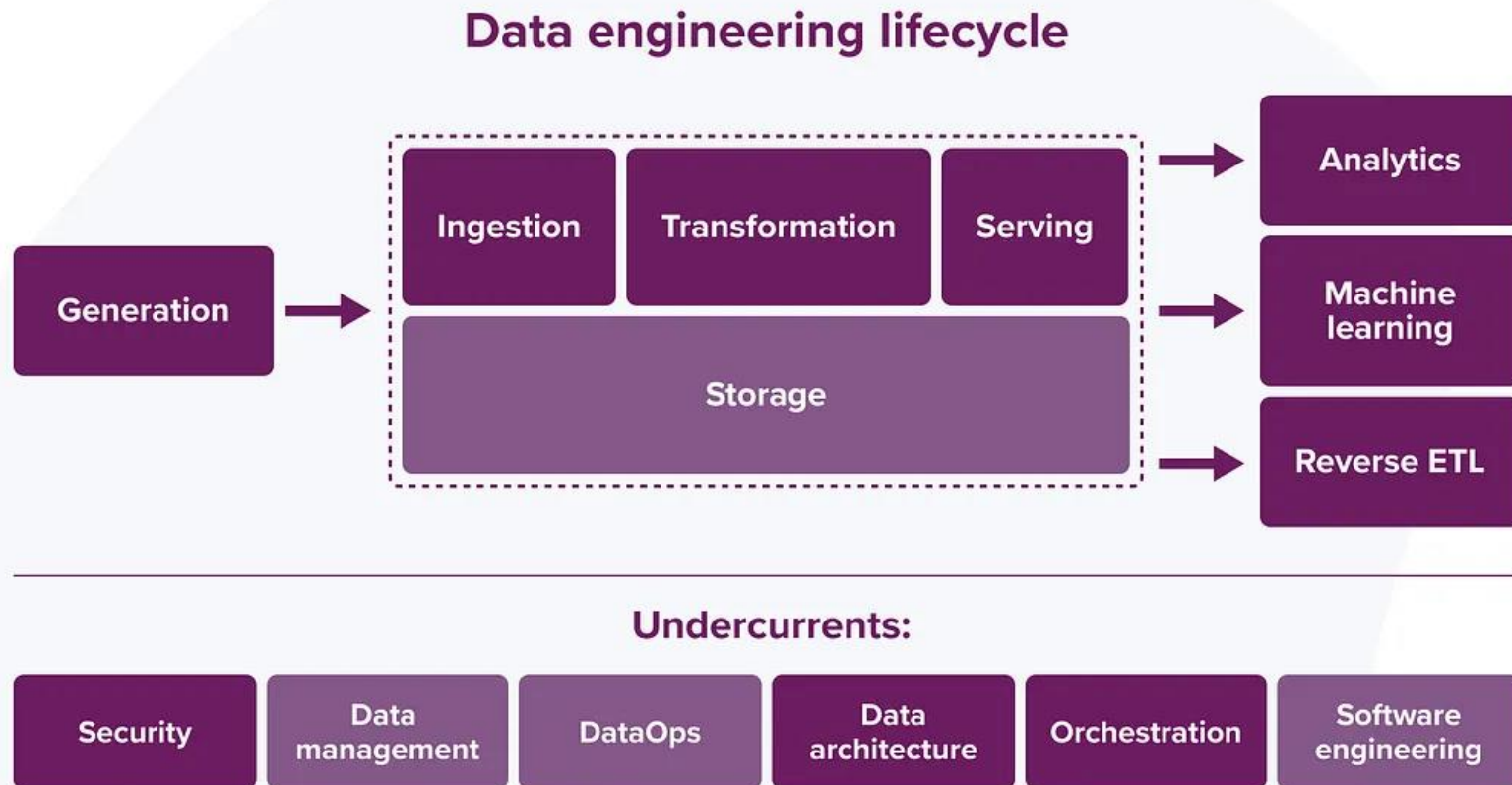
9. The Data Engineering Lifecycle

➤ **Stages:**

1. Data Ingestion: Collecting raw data.
2. Data Transformation: Cleaning and organizing the data.
3. Data Storage: Storing data in databases or data lakes.
4. Data Analysis: Passing data to analysts or scientists.

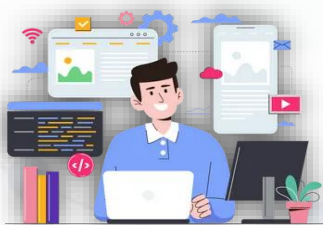
➤ **Example:** Uber's use of data to optimize ride routes.

Data Engineering Lifecycle



10. Collaboration in Data Engineering

- **Data Engineers:** Build infrastructure.
- **Data Analysts:** Analyze data for insights.
- **Data Scientists:** Develop models to predict future trends.



11. Summary

➤ Key Takeaway:

- Data engineers are responsible for the flow of data in organizations.
- ETL, pipelines, and big data are core concepts in data engineering.
- Popular tools includes Hadoop, Spark, Kafka, and Airflow.
- The lifecycle of data engineering involves ingestion, transformation, storage, and analysis.



Question & Answer

